# Predictors of Mental Health Service Utilization Among Young Adult Males

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## I. Synopsis

The National Survey on Drug Use and Health found that approximately 49.4% of young adult females who reported any mental illness sought out mental health services, while only 32.9% of young adult males sought care (SAMHSA, 2024). This highlights a mental health trend that remains prudent in our society in which there is a marked gender gap in treatment access and outreach. Females are consistently more likely to access treatment than their male counterparts despite the fact that there has been an upward trend of mental health issues post-COVID. This begs the question, why? Why aren't more young adult males seeking care and can we identify these disparities to patch them with proper care and policy? This led our team to ultimately come up with the question: Among U.S. young men aged 18–25 who report experiencing mental health challenges, what demographic, socioeconomic, and geographic factors predict whether they utilize professional mental health services? We hypothesize that socioeconomic and demographic factors such as income level, race, and education shape both the likelihood of young men reporting mental health struggles and seeking professional help. We predict that young men who live in more politically conservative states will be less likely to report mental health challenges and utilize services due to social stigma, cultural norms, and reduced access to care.

## II. Background & Question

As touched on, mental health challenges among young adults are at historic highs, with the COVID-19 pandemic further exacerbating depression and anxiety rates. In 2021 alone, the NSDUH found that 41.7% of young adults aged 18-25 reported experiencing mental illness in the past year - making this the most affected age group in the U.S (SAMHSA, 2024). When broken down by gender, we see disparities in service utilization emerge - an alarming reality. Men consisting of the young adult category are less likely to seek out mental health services,

highlighting a barrier: although young men face high rates of mental health challenges, they are less likely to access care. By focusing specifically on young men, we seek to address a critical question: what factors most strongly predict whether they seek help? Understanding these predictors can guide targeted interventions to reduce barriers while also improving mental health outcomes for a population that is underserved.

## III. Data

For this project, we used the 2021 data from the Medical Expenditure Panel Survey (MEPS), including four main files: the Full-Year Consolidated Data File (HC-233), Jobs File (HC-227), Office-Based Visits File (HC-229G), and Longitudinal Weight File (HC-226). We also used The National Survey on Drug Use and Health. These files were downloaded from the official websites in .CSV and .DAT formats and no APIs or scraping methods were needed. The MEPS HC-233: 2021 Full Year Consolidated Data File provided us with demographic information (age, race, education, income, poverty level, insurance coverage, and self-perceived mental health status). This gave our population structure and predictor variables. The MEPS HC-226: Panel 23 Longitudinal File was used in order to track individuals across time and help identify their mental health utilization. MEPS HC-227: 2021 Office-Based Medical Provider Visits was beneficial for us to identify actual usage of mental health services as this data set includes a ‘reason for visit’ and provider type. The National Survey on Drug Use and Health (NSDUH) gave us a National-level benchmark rate and allowed for comparison to our MEPS-based sample. We used the data from 2021 as this is the most recent full-year MEPS data which offered information for young men ages 18-25. It also captured a unique perspective in history, post COVID and allowed us to identify its impact, especially as mental health issues surged among young adults.

To begin cleaning our data we filtered the 2021 MEPS HC-233 Data file in order to isolate our target population of young adult men ages 18-25. We selected a key group of variables which included age, sex, race, education, poverty status, total income, insurance coverage, census region, and self-reported mental health status. We also re-coded numeric categorical variables into meaningful labels (ex. from 1 or 2 to Male or Female). Alongside, we used the variable MNHLTH53 in the HC-233 dataset to quantify mental health. It gives us a 5-point scale (1=Excellent, 2=Very Good, 3=Good, 4=Fair, and 5=Poor). In order to enrich our data from the HC-233 file, we merged in supplementary data from the HC-277 Prescribed Medicine data set and HC-299G Office-Based Medical Provider Visits. We used the unique person identifier (DUPERSID) to perform these merges. From the H227 file we selected industry, occupation, and employer-sponsored insurance to help us assess potential socioeconomic determinants of mental health service access. We utilized the H229G file to select variables to indicate whether the visit was for mental health, whether a provider was seen, or a medication prescribed. These were then filtered to isolate mental-health related visits. Both inner joins and left joins were used either to uncover individuals and their complete records across data sets or to preserve individuals from our base HC-233 file, even if they did not appear in the others. The utilization of R helped us to then perform a visual exploratory data analysis.

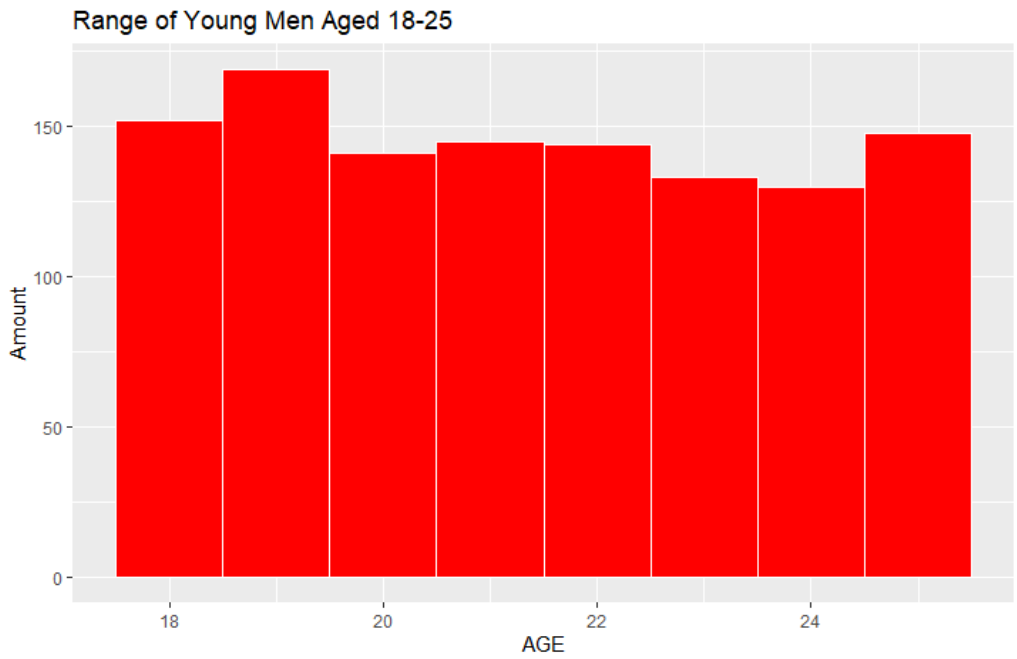



*Figure 1: Distribution of Age Among Young Men (18-25) within Sample*

Our first bar chart, generated through R, displayed the count of individuals for each age between 18 and 25. The distribution among men is relatively uniform, with a slight spike among age 19 and a slight decrease at age 24.

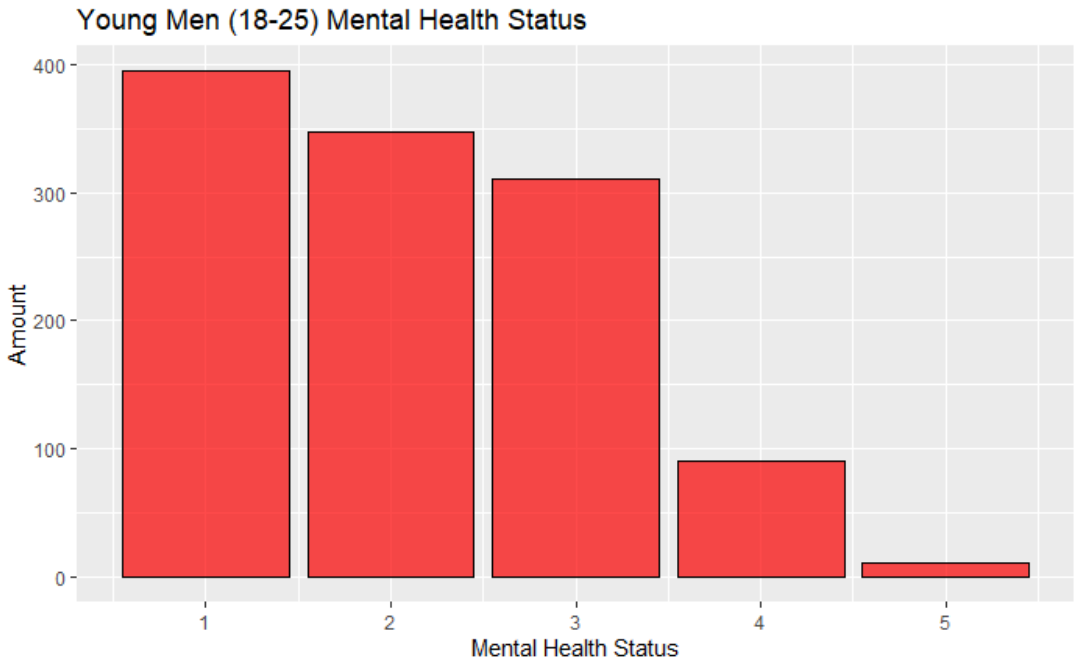



*Figure 2: Reported Mental Health Status Among Young Men*

Our second bar chart revealed that most young men within our data sample rated their mental health in the lower severity categories at a '1' as opposed to an extreme severity scored by '5'. We observed that very few respondents in our population selected 4 or 5, suggesting a skew towards a more favorable mental health self-perception.

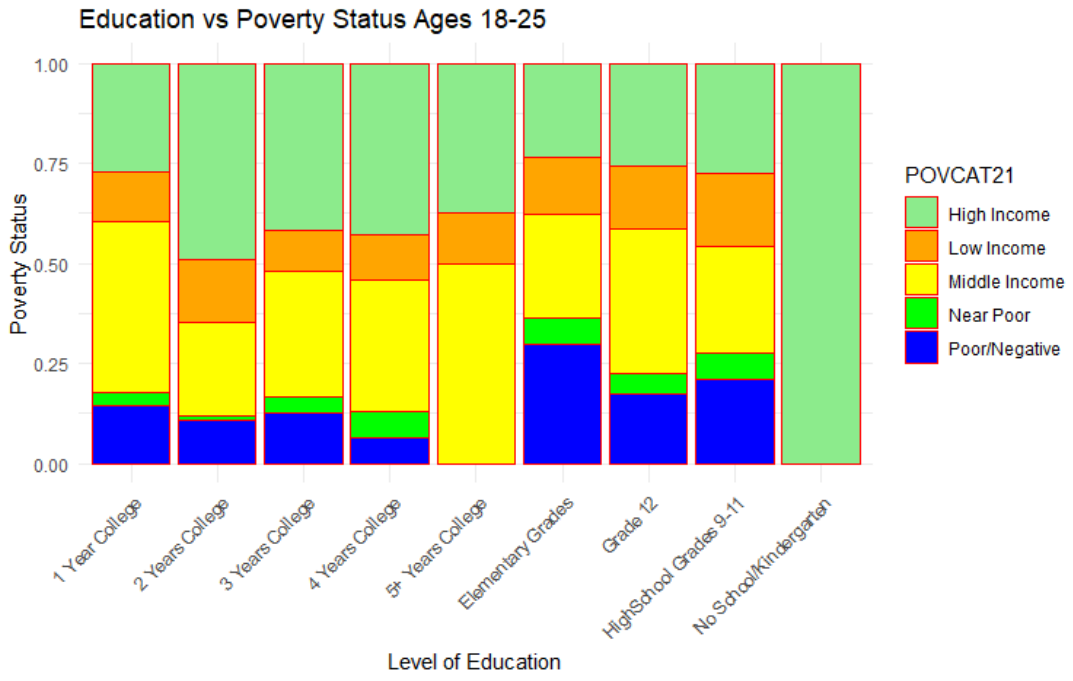



*Figure 3: Distribution of Poverty Status by Education Level Among Young Men*

This stacked bar chart offered insight on how poverty status varied across education levels. We observed that higher levels of education, specifically 4+ years of college were associated with a greater proportion of individuals in the high-income category. In contrast, those with lower education levels (no GED or elementary education) were more likely to fall into near poor/poor poverty levels. This graph helped shape the validity of our hypothesis as we were able to explore the relationship between education and its association with socioeconomic status and ultimately its influence on mental health service utilization.

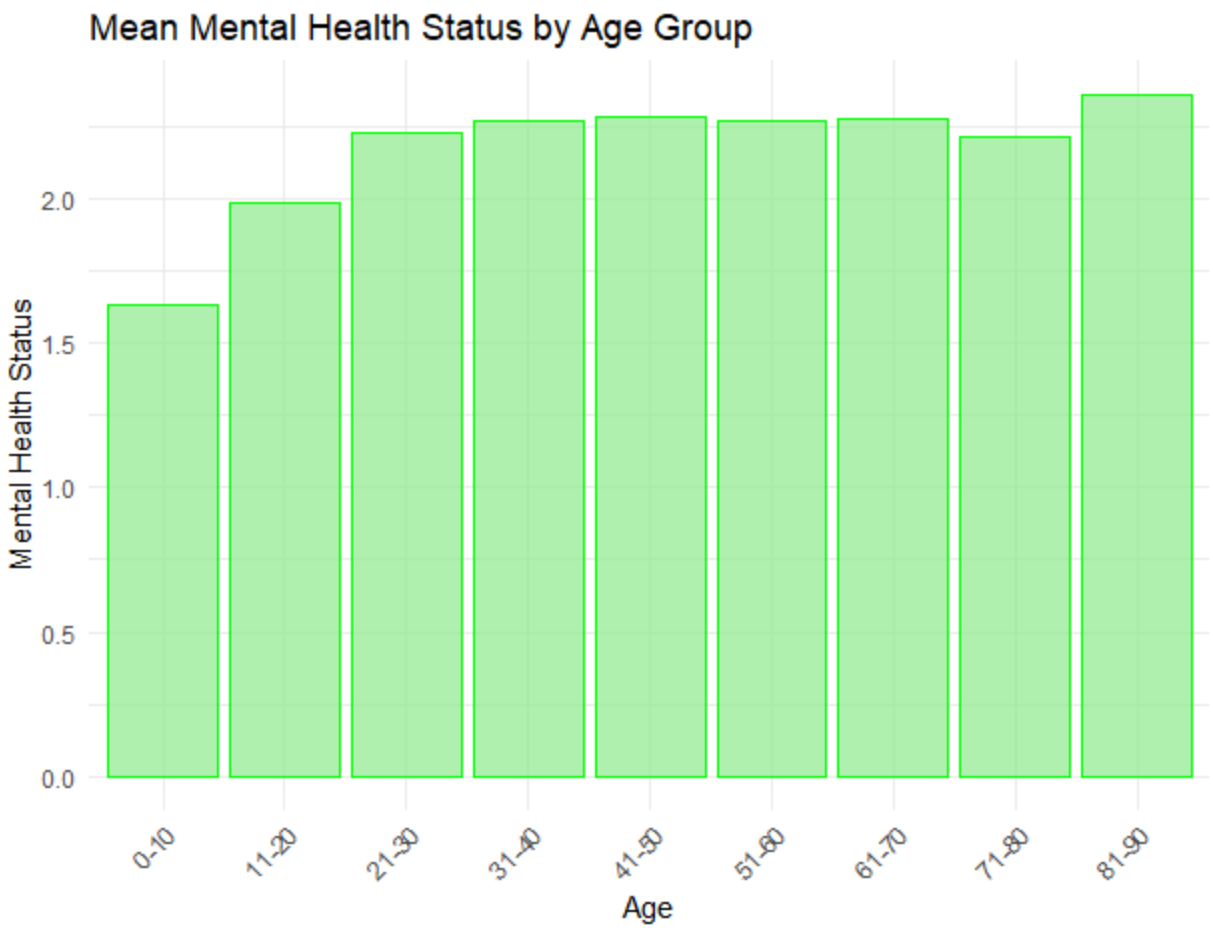
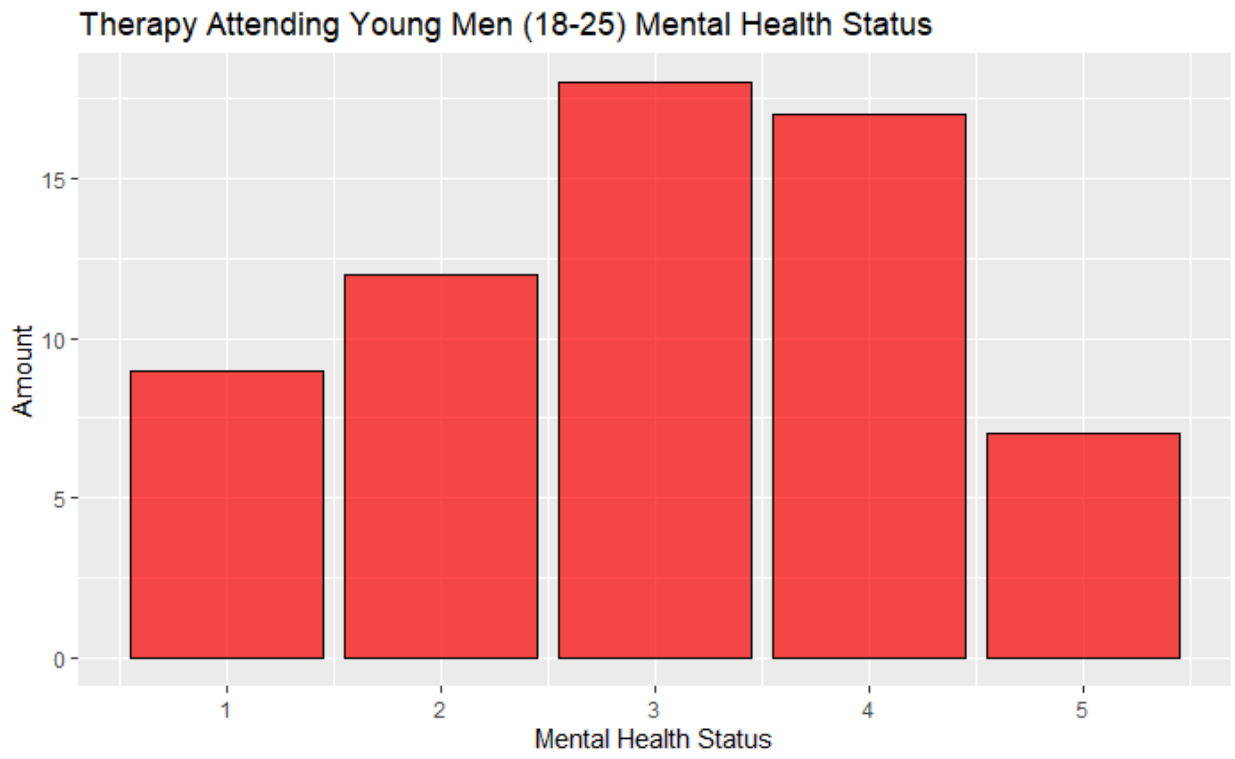



*Figure 4: Mean Mental Health Status by Age Group among Men*

This chart aided us in interpreting how mental health status varied by age group across the full age spectrum and allowed us to develop contextual insights for comparison. Younger individuals tend to report better mental health (between 1.5 and 2.0) with a gradual increase (above 2.0) as age rises. This visual suggested that there are increased mental health challenges later in life with a steady level through mid-life. Perhaps if we target mental health support in early ages, we could see healthy levels of mental health status later in life.


Therapy Attending Young Men (18-25) Mental Health Status
Amount
0
5
10
15
1
2
3
4
5
Mental Health Status

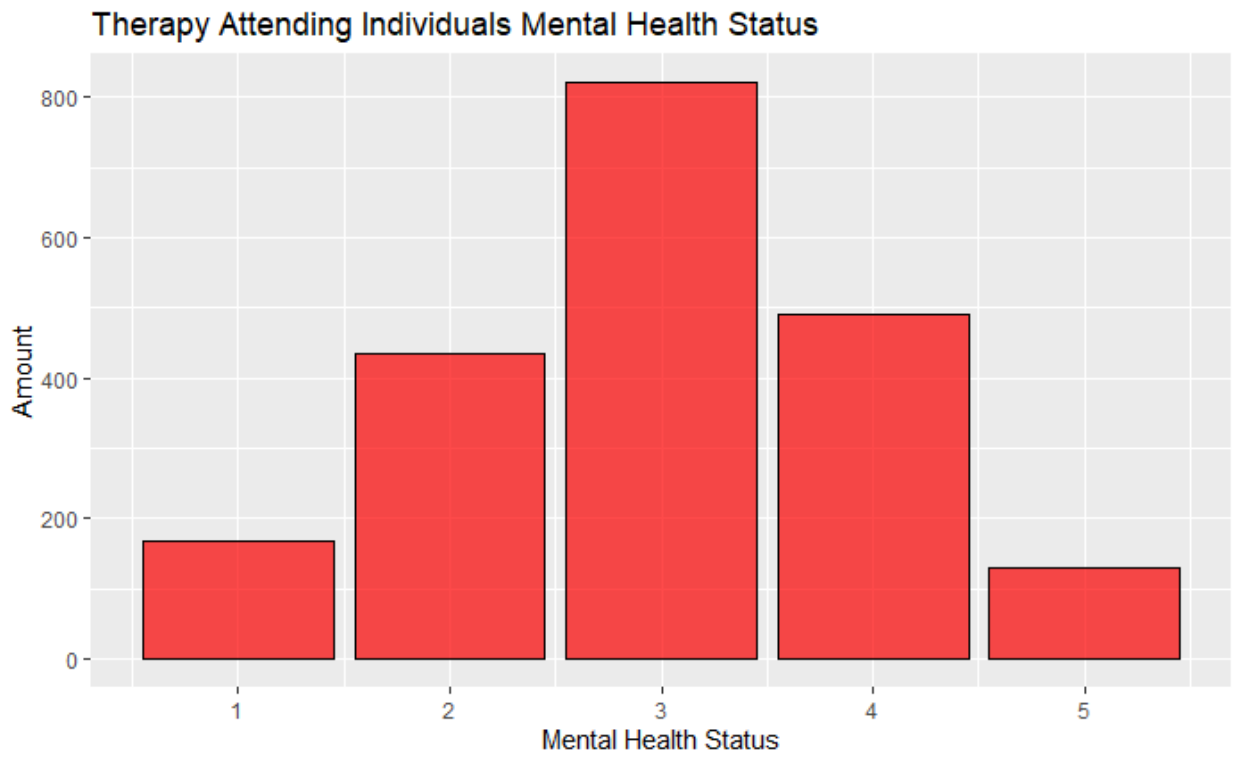

Therapy Attending Individuals Mental Health Status
Amount
0
200
400
600
800
1
2
3
4
5
Mental Health Status

*Figure 5a and 5b: Comparison of Mental Health Status Distribution among therapy attending young men (top) vs. individuals (bottom)*

Figure 5a displayed the self-reported mental health status of young men who reported attending therapy. This distribution skewed towards moderate (3) and favorable (4) mental health status, with fewer individuals reporting more severe (1) or best (5) scores. Figure 5b showed the broader population, including women who had attended therapy. This clustered towards the middle of the scale with 3 being again the most common rating. It is important to note that the amount of individuals reporting is higher.

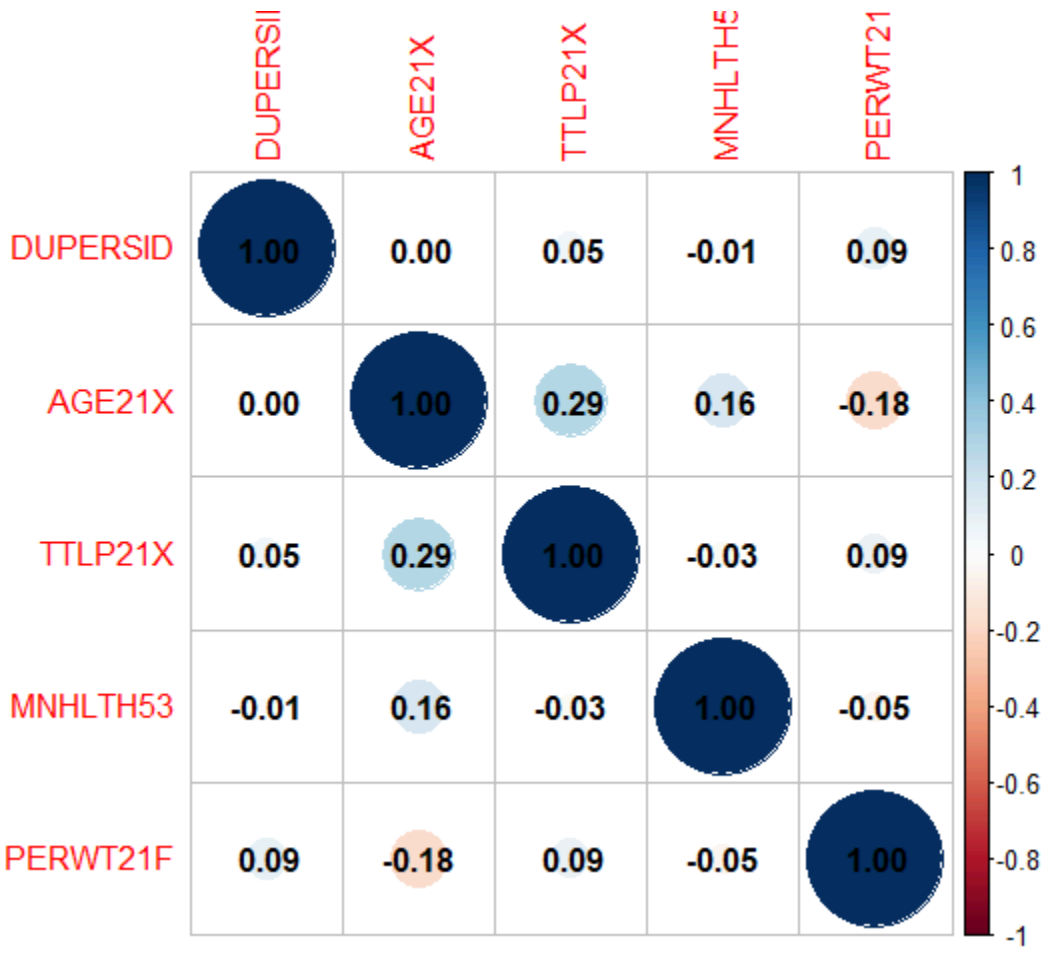


*Figure 6: Correlation Matrix of Key Variables*

This correlation matrix illustrated the relationships among age, total personal income, perceived mental health, and weight. We observed a slight positive relationship between age and income and a mild negative relationship between age and weight. These findings provided us with preliminary insight into how these variables might interact in later modeling steps.

| | POVCAT21 | n | percentage |
|---|---|---|---|
| 1 | High Income | 10063 | 35.513128 |
| 2 | Low Income | 3987 | 14.070440 |
| 3 | Middle Income | 7858 | 27.731508 |
| 4 | Near Poor | 1486 | 5.244212 |
| 5 | Poor/Negative | 4942 | 17.440711 |

*Figure 7: Categorical Variable Table for Poverty Status*

This categorical variable table for poverty status showed us the frequency count and percentage for each level of poverty - ranging from (1) High Income to (5) Poor/Negative Income.

| Skewness_MNHLTH53 | Kurtosis_MNHLTH53 | Sd_MNHLTH53 | Median_MNHLTH53 | Mean_MNHLTH53 | Min_MNHLTH53 | Max_MNHLTH53 |
|---|---|---|---|---|---|---|
| -2.088801 | 15.28461 | 1.26343 | 2 | 2.141093 | -8 | 5 |

*Figure 8: Continuous Variable table for Mental Health Status*

Figure 8 presented us with the summary statistic for our self-reported mental health status variable. The mean mental health score was 2.14 and the median was 2 which suggested that the majority of responses fell in the 'Very Good' to 'Good' range. The distribution was left-skewed with a high kurtosis of 15.28. This indicated a sharp peak and heavy tails - potentially due to a large number of favorable self-reports and few of the alternatives.

## IV. Models

In order to evaluate predictive performance, a training/test split was necessary using the initial_split() function in R. This function enabled us to randomly partition the dataset into two subsets - a training set (75%) and a test set (25%). We also ensured that the proportion of individuals who reported using mental health services was balanced among both subsets. This was prudent to avoid any class imbalance. This lays the foundation for our learning models.

Feature engineering included creating age brackets along with a variable to represent the number of visits to a mental health professional per person. Alongside, we explored the possibility of binary transformation, specifically with grouping perceived mental health scores (1-3=good and 4-5=poor) in an attempt to improve prediction performance.

Firstly a Generalized Linear Model (GLM) was used to help shape our approach to our Linear Discriminant Analysis (LDA). By noting what variables weren't significant to the GLM, we were able to hone our formal LDA. The LDA was then applied to begin exploring supervised learning methods which distinguish between individuals who did and did not utilize mental health services. The honed model performed well without the fat, performing just .06% less well than a model with all variables in our ordinal data and .02% less well with the binary transformation.

Moving on to supervised learning algorithms, we explored the NaiveBayes, Random Forest and XGBoost. These were chosen due to their ability to handle diverse data types, rank feature importance, and capture nonlinearities. Additionally, Random Forest and XGBoost are adept for multicollinearity and do not require transformation or predictors, proving particularly useful for this dataset. Other models considered briefly were Linear Models and PCR.

When exploring the Random Forest models, we first approach treating mental health as a continuous outcome. Any unnecessary variables were removed, keeping only predictors with strong relevance (Race, education, poverty, insurance coverage, income, and region).

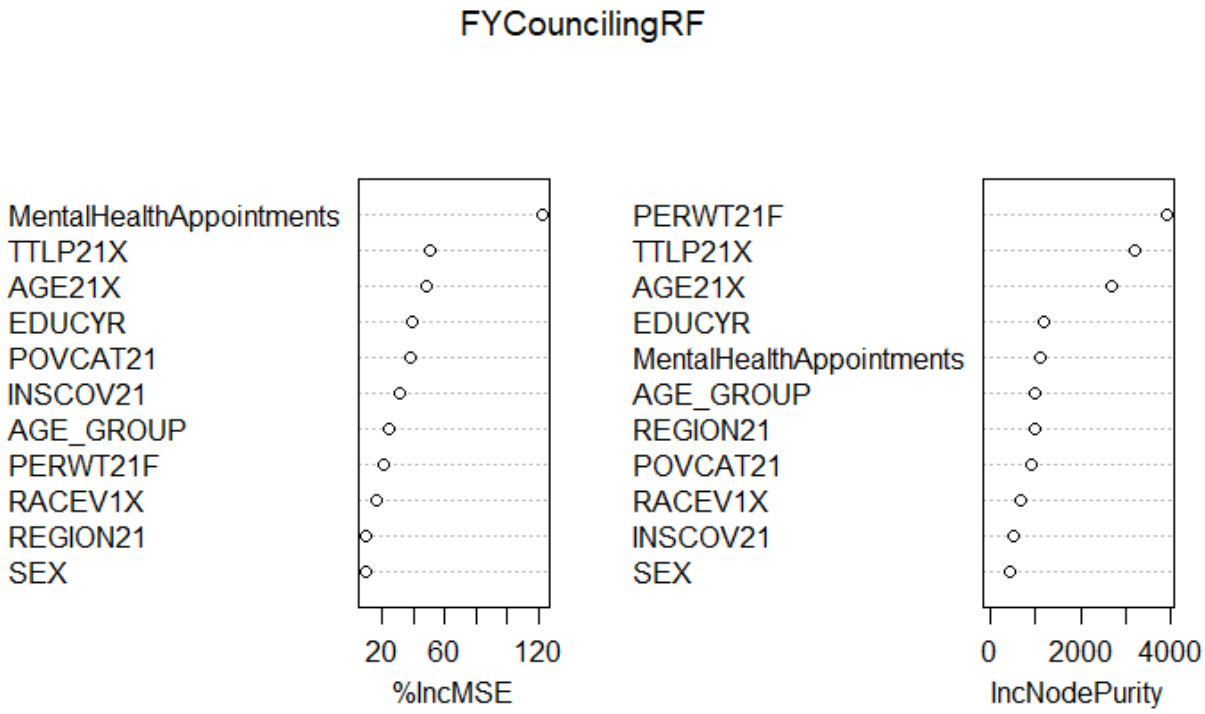



*Figure 1: Variable Importance Plot for Model 1 (We Used to eliminate variables from Model 2)*

In order to refine our predictive model, we generated a variable importance plot from the initial Random Forest model, using two metrics - %IncMSE and IncNodePurity. Our results yielded the top predictors to be the following: Total family income (TTLP21X), Age (AGE21X),

weight (PERWT21F), and education years (EDUCYR). This allows us to filter and exclude variables with minimal contribution to our models accuracy, leading to reduced noise. This helped us create model 2 which was trained on the most informative features.

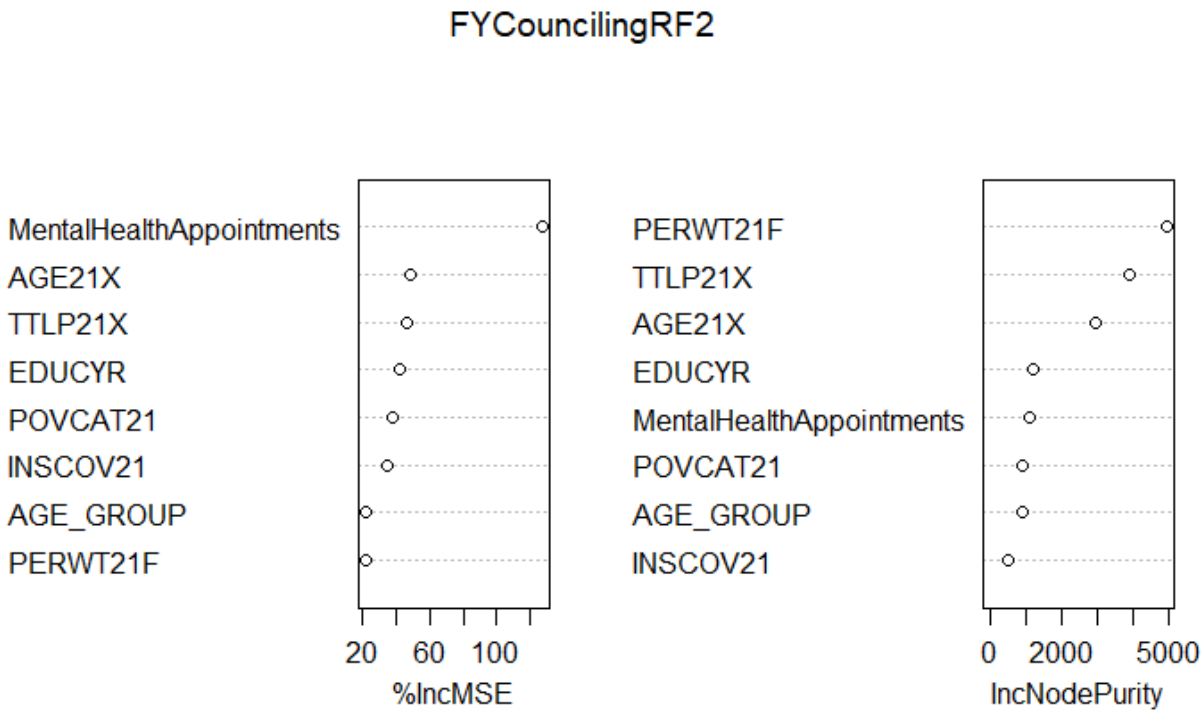



*Figure 2: Variable Importance Plot for Model 2*

For our first model, we used our most influential predictors - suggesting that demographic and socioeconomic indicators such as age, total income, and insurance coverage may play key roles in predicting mental health service utilization. The model was then evaluated

using accuracy, AUC-ROC, and precision.

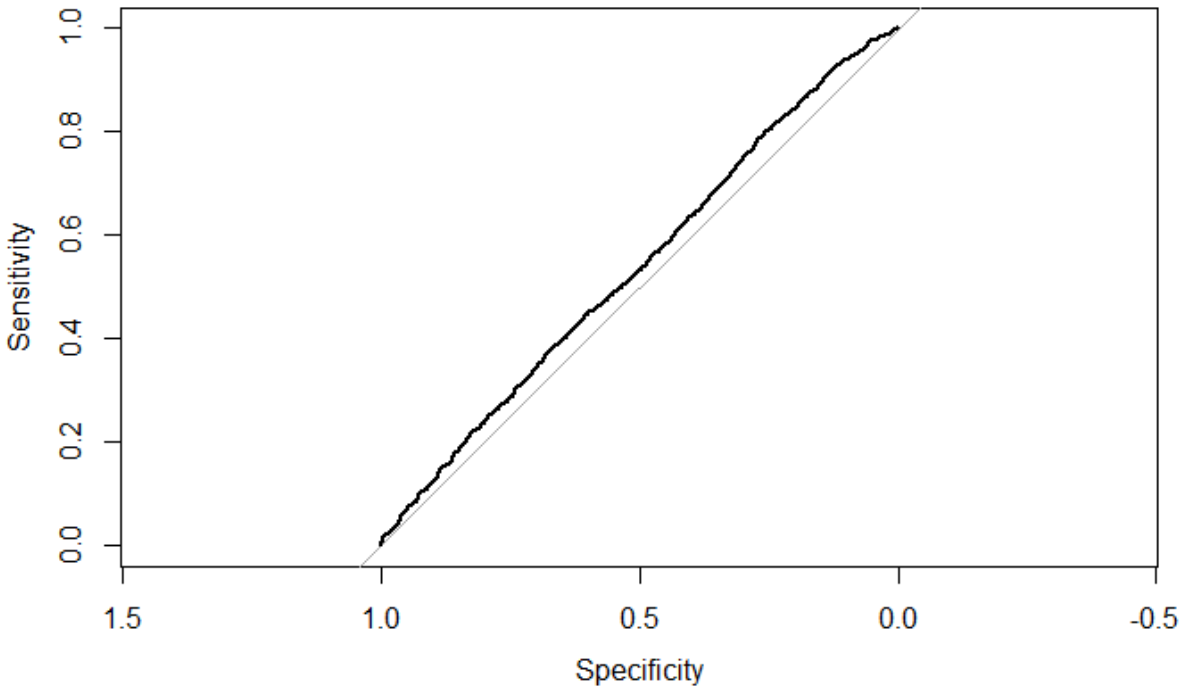



*Figure 3: ROC curve for Model 2. AUC = .5362*

Our AUC value is 0.5362 which is only slightly above the baseline of 0.5. This is a notably low AUC, suggesting this model is having difficulty distinguishing between positive and negative classes. But to give this model credit, the imbalance of the negative and positive classes creates an uphill battle for our algorithm. No matter the difficulty, this analysis yields the result that this model's predictive performance is not strong.

Our second approach was to use the binary classifier we set up for mental health earlier. This model's performance is likely a result of the feature selection process that was initiated in Model 1 where we removed variables that scored low on importance. This allowed Model 3 to have a streamlined feature set, which was further exacted by converting mental health utilization into a binary variable (Yes or No). We considered that a categorical output would be better suited

for this algorithm.  This combination likely led to a higher AUC for our second model. However, we are still aware of the potential limitations that come with this model, indicated by a high false negative rate for predicting ‘Poor’ health outcomes. We recognize that although the model more sufficiently distinguishes between classes overall, it does not give us a full look into at-risk individuals.

For assumptions, we used the variable importance plot to help visualize what variables were not carrying their weight in MSE and Node Purity. Additionally, we used the confusion matrix from the second as we noticed numerous false positives due to having a significant amount of data claiming good mental data as opposed to bad. We would like to explore a more accurate model for detecting poor health successfully as this will be vital in supporting our hypothesis.

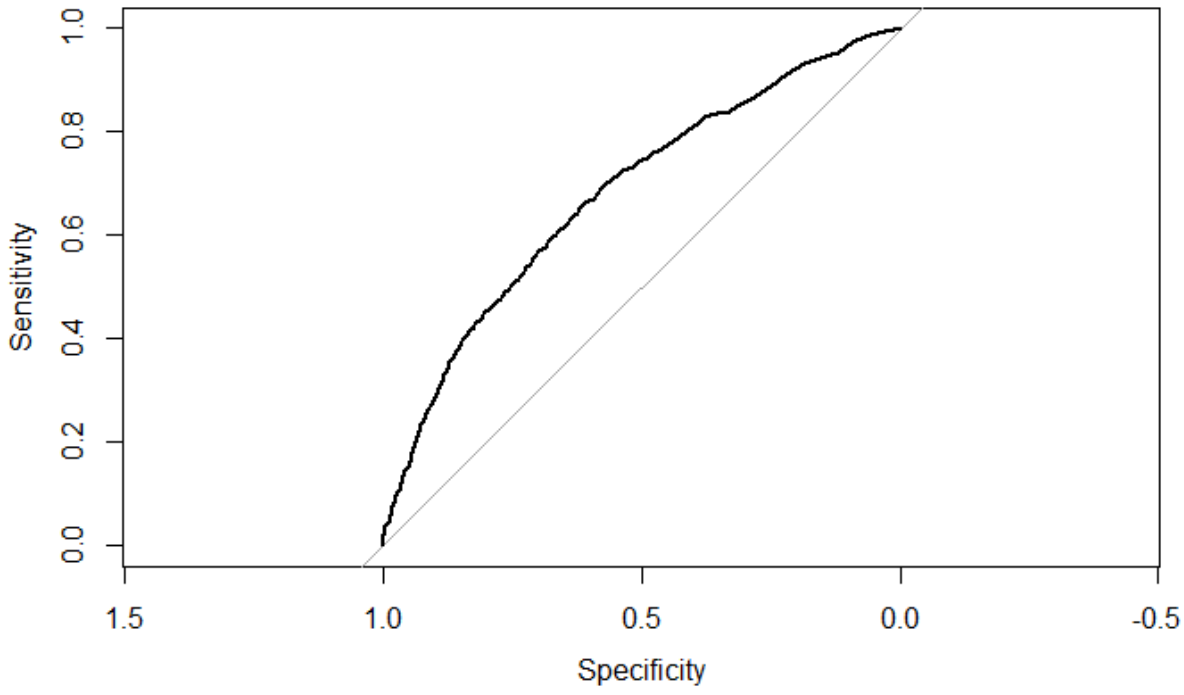



*Figure 4: ROC for Model 3 AUC = .6803*

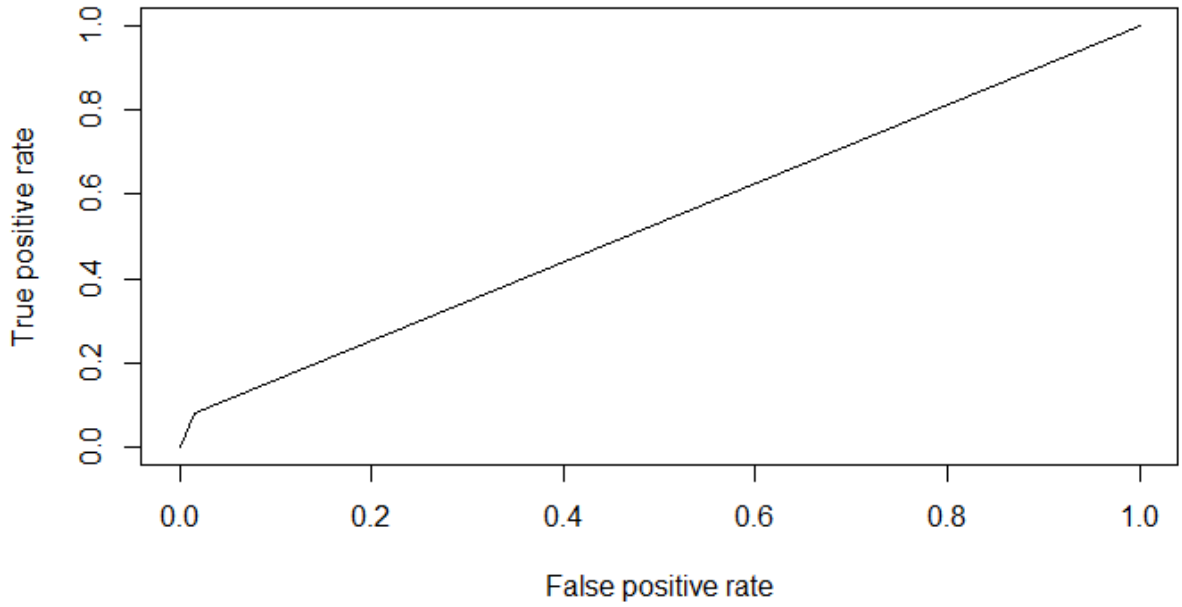
Model 3 has a noticeably higher AUC score which we will further investigate.

```
Confusion matrix:
      Good Poor class.error
Good 17003  266  0.01540332
Poor  1896  148  0.92759295
```

*Figure 5: Confusion Matrix*

Random Forest Models are naturally resistant to overfitting. Although we did not do formal cross-validation, out-of-bag error rates provided a built-in validation check.

Exploring the NaiveBayes model next, this model scored the best in terms of accuracy when using the binary variable, beating our second most accurate model by a slim margin. The issue with this model was that it produced unsatisfactory ROC results. The area under this curve was only 0.5329.



Nevertheless, we deemed the large accuracy was enough to warrant exploring the model after it's been cross validated. After settling for a ten fold cross validation we found the model's accuracy did not shift much in terms of accuracy or AUC.

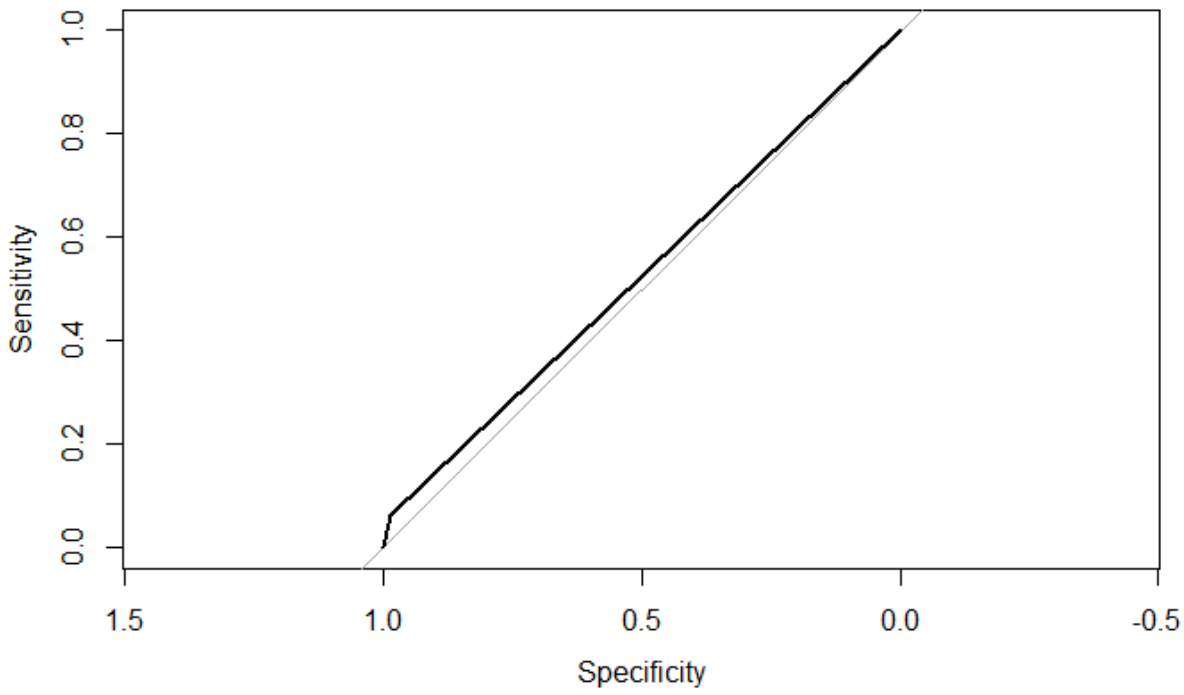



*Figure 7: ROC for Model 3 AUC = .5246*

The last significant addition and our final model was SMOTE, so we could address class imbalance in the mental health outcome variable. For this analysis we decided to stick with the binary factor variable so we could have more poor representation in our confusion matrix. Using this balanced training set, we tuned and rebuilt a Random Forest Model. Hyperparameters explored included:

- mtry = 4, ntree ≈ 1000, and importance = TRUE

We also tried a "trimmed" feature set by removing low-importance predictors which degraded test MSE, so we reverted to the fuller set. In summary we:

- Reframed the target to a binary classification (MNHLTH53 = Good vs Poor)
- Fit an RF classifier with mtry = 4
- Kept all predictors except the identifier (DUPERSID)

```
smote_class Good Poor
       Good 5411  540
       Poor  370  117
```

*Figure 8: Confusion matrix for SMOTE data*

For threshold selection we used the ROC curve to choose a cutoff and we confirmed out-of-bag (OOB) error against the held-out test set in order to check for overfitting. Ultimately, choices were made to keep the search simple/replicable early (using 1-D grid) and to align tuning with the study question of detecting "Poor" mental health.

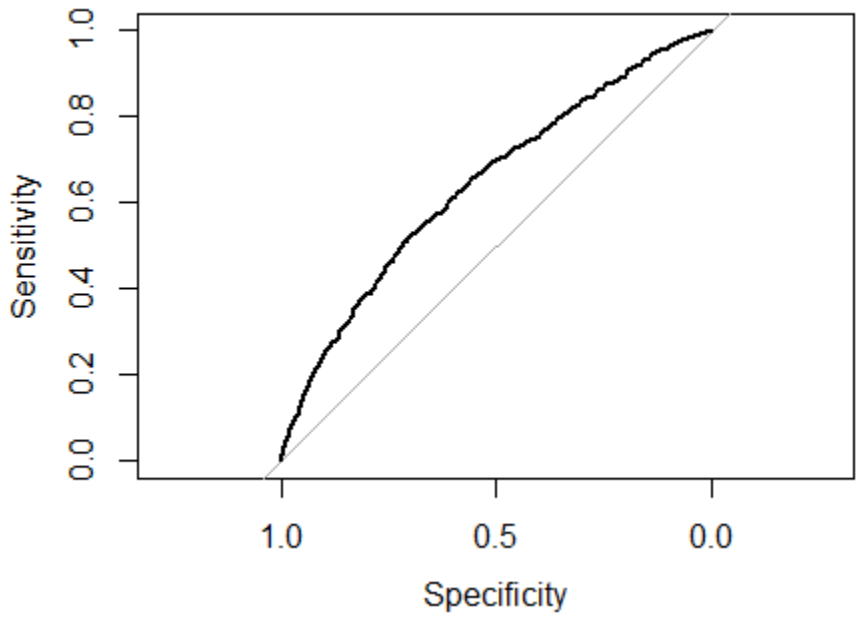
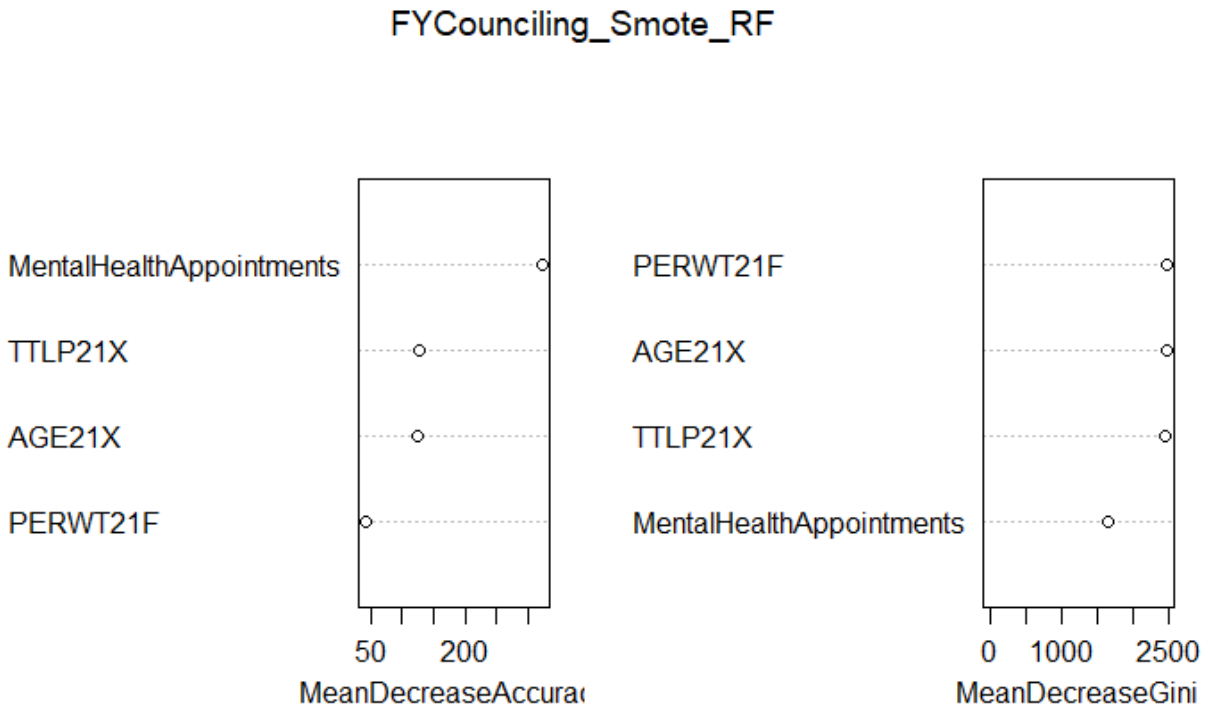



*Figure 9: Random Forest - SMOTE*

```
> auc(ROC_FYCounciling)
Area under the curve: 0.6441
```

*Figure 10: AUC - SMOTE*

While this model gives us the best performance when detecting accurate Poor mental health ratings, this model does have a minor flaw. Since SMOTE only processes numeric variables, our mostly categorical data doesn't see much use. However each numerical piece of data plays a crucial role in our model. Each variable was significant in either Mean Decrease Accuracy or Mean Decrease Gini.




| | Good | Poor | MeanDecreaseAccuracy | MeanDecreaseGini |
|---|---|---|---|---|
| AGE21X | 73.46935 | 103.87624 | 124.3181 | 2463.170 |
| TTLP21X | 66.94547 | 165.99689 | 127.9867 | 2451.864 |
| PERWT21F | -4.00497 | 81.85566 | 40.9907 | 2479.653 |
| MentalHealthAppointments | 232.45577 | 254.11176 | 324.3258 | 1644.293 |

*Figure 11: Variable Importance Plot for SMOTE*

## V. Conclusions

In conclusion, the question at hand aimed to solve what demographic, socioeconomic, and geographic factors predict whether young men aged 18-25 within the U.S. who have reported experiencing mental health challenges utilize professional mental health services. Age and total family income proved to play a significant role in predicting the utilization of mental health services. This was proven through data cleaning, feature engineering, and model creation. The models used were LDA, GLM, XGBoost, Naive Bayes, Random Forest, and SMOTE. These findings directly tie to the hypothesis at hand, and are linked with the discoveries from the National Survey on Drug Use and Health. In the end, the Random Forest SMOTE model most accurately showcased the impact of the above features on young men in relation to mental health services. Although certain categorial factors were forgoed, the overall idea persisted. It is important to note that the additional Random Forest models tested highlighted education level and insurance coverage as important demographic and socioeconomic factors as well. Young men can experience very detrimental mental health challenges, and it's integral that those without the insurance, education, and income are still being helped.

## VI. Discussions and Next Steps

The main takeaway from this academic experiment was that total family income and age are the main socioeconomic and demographic factors that influence young men seeking mental health resources. This can be best viewed through figures 9 and 11. The Random Forest SMOTE model, though having an AUC of 0.6441, gives us the best performance when detecting accurate poor mental health ratings. This is a vital source of information as we deem which individuals and characteristics are relevant to the question at hand. This model did see restraints such as class imbalance and a lack of categorical variables. This is why we choose to also take into account our other Random Forest models which highlighted the importance of education and insurance in regard to mental health services. It is recommended that future analysts of this

question seek additional data sources, models that incorporate both numerical and categorical variables, and information pertaining further into geographic and cultural influences. Further recommendations to policymakers include expanding the accessibility of health insurance and creating an open conversation on mental health specifically in relation to men.

## VII. Code Availability

Our GitHub link can be found here:

https://github.com/duganti/6311-Group-4-Capstone/tree/main.

**Appendix: Data Dictionary**

| Variable | Meaning |
|---|---|
| DUPERSID | Person ID |
| AGE21X | Age in 2021 |
| SEX | Gender |
| RACEV1X | Race |
| EDUCYR | Years of Education completed |
| POVCAT21 | Poverty Category |
| TTLP21X | Total Personal Income |
| INSCOV21 | Insurance Coverage Status |
| REGION21 | U.S Region |
| MNHLTH53 | Perceived Mental Health |
| PERWT21F | Weight |
| INDCODEX | Industry Code |
| OCCCODEX | Occupation Code |
| EMPLINS | Health Insurance through employer |
| VSTCTGRY | Visit Category |

Center for Behavioral Health Statistics and Quality. (2024). Differences in past year mental health among young adults aged 18 to 25 (NSDUH Data Brief No. 1). Substance Abuse and Mental Health Services Administration. https://www.samhsa.gov/data/sites/default/files/reports/rpt44475/2021-nsduh-data-brief-1.pdf